

Green Room Air Conditioner Specifications (Draft)

Background

As a part of the ongoing support to UNEP and the Ministry of Finance (MoF), a framework for sustainable public procurement (SPP) with a special focus on room air conditioners (RAC) was developed under the MAITREE program. This is being reviewed by the Task Force on Sustainable Public Procurement constituted by the Department of Expenditure, Ministry of Finance.

Continuing the partnership with UNEP, support for working with Government e-Marketplace (GeM) for launching SPP is now being provided under the MAITREE program. One of the key outcomes will be the launch of the first-ever green air-conditioner procurement for India.

GeM would like to introduce a category for “green products” starting with room air conditioners. The specifications for the “green air conditioner” are based on the SPP criteria outlined in partnership with UNEP. While we continue to work on efforts of SPP integration with GeM, this is a separate exercise that is focused on a separate product category, for those seeking “green air conditioners”. Hence the following aspects have been considered while suggesting the specifications for “green air conditioner”:

1. All specifications outlined under core and comprehensive criteria have been included from the SPP report. However, some of the comprehensive criteria have been included as qualitative measures.
2. Specifications related to energy consumption and refrigerants have been made more stringent based on the current market assessment and to promote a few innovative RAC manufacturers.

Product Specifications (Room Air Conditioners till 1.5 TR cooling capacity)

All “Green Air Conditioners” will be variable speed (inverter)-based. Fixed speed RACs will not qualify.

<i>Product Specifications for Room Air Conditioners</i>	
<i>Parameter</i>	<i>Specifications</i>
Safety and Performance	Air conditioner shall conform to the requirements for quality, safety and performance prescribed in IS 1391 Revised /IEC 60335-2-40 (under preparation) and all other requirements specified in this standard.
Product Noise	Air conditioner shall conform to the noise levels as notified under the Environment (Protection) Act, 1986 from time-to-time but not more than the limits specified under the standard issued by BIS (IS 1391 Revised).
Energy Consumption	The ISEER shall be not less than the value prescribed below as per norms specified by the Bureau of Energy Efficiency (BEE) from time to time. • Up to 3517 W (upto 1 TR) – ISEER 5.8 • (3517 W to 5280 W) (1 to 1.5 TR) – ISEER 5.33
Refrigerants	The refrigerant should have Zero ODP. The Global warming potential (GWP) not exceeding 700 (100 years) ¹² is required until a specific directive is issued by MoEFCC.

<i>Product Specifications for Room Air Conditioners</i>	
<i>Parameter</i>	<i>Specifications</i>
Recycled Plastic Components	Product shall be designed to promote recycling, by utilising recyclable plastic components at least 80 percent by weight of plastic components in product ¹
Paint	Paints used in the product shall not contain heavy metals or their compounds include mercury (Hg), lead (Pb), cadmium (Cd) and hexavalent chromium (Cr).
Packaging	The air conditioner shall be packed in such packages, which are made of recycled or biodegradable materials. Plastic packaging shall not contain halogenated hydrocarbon.
Green Clauses	Take-back or buy-back option is available with the manufacturer.

<i>Organizational Specifications for RAC Manufacturers</i>	
<i>Parameter</i>	<i>Specifications</i>
Hazardous substance management	The manufacturers should abide by the regulations listed under the Act regarding hazardous waste; including, not limited to, the following²: • The Environment (Protection) Rules, • Hazardous Waste (Management, Handling & Transboundary Movement) Rules, • E-Waste (Management) Amendment Rules, • Plastic Waste (Management and Handling) Rules, • Solid Waste (Management) Rules • The Plastics (Manufacture, Usage and Waste Management) Rules • The Recycled Plastics Manufacture and Usage Rules, • Batteries (Management and Handling) Rules • The Manufacture, Storage and Import of Hazardous Chemical Rules. The Acts and Rules made thereabove shall be read including all amendments to date.
Noise Pollution	The manufacturing facilities shall comply with the noise standards for industrial facilities, as specified within the provisions of the Act and Noise pollution (Regulation and Control) Rules. Additionally, all such facilities should take measures for the abatement of noise, including noise emanating from the sound producing equipment or instruments, and should ensure that existing noise levels do not exceed ambient air quality standards specified. All planned developmental activity related to industrial manufacturing or distribution of manufactured products should take into consideration noise pollution aspects and should avoid noise menace. The recommended ambient noise levels shall always be adhered to by cooling appliance manufacturing facilities.
Ozone Depletion	The complete list of ozone-depleting substances is specified on the website of the Ministry of Environment, Forests, and Climate Change (MoEFCC), and includes CFCs, Halons, Carbon Tetrachloride, Methyl Chloroform, HCFCs, BFCs, HBFCs, and Methyl Bromide. No manufacturing facility shall employ equipment that releases ozone depleting substances, and all existing equipment should be in the process of phasing out ozone-depleting substances.

¹ Reference: Korea Eco-Label. Air Condition: Korea Environmental Industry & Technology Institute. 2011; and Thailand Environment Institute - Green Label Product: Room Air Conditioner (TGL- 7- R3 - 14)

² Information on Rules is available on MoEFCC website. The Acts and Rules made thereunder shall be read including all amendments till date.

<i>Organizational Specifications for RAC Manufacturers</i>	
<i>Parameter</i>	<i>Specifications</i>
	No manufacturing facility shall export or import ozone-depleting substances, equipment, or instruments to any country. No manufacturing facility or any associated person(s), shall sell, stock, or exhibit for local or international sales, any ozone-depleting substance, equipment, or instrument. No manufacturing facility or any associated person(s), shall establish, expand, or invest in, ozone-depleting substances, equipment, or instruments.
Environmental Management System (EMS) Certification	Compliance with ISO 14001 (EMS – Environmental Management System)
Corporate Social Responsibility	Comply with Corporate Social Responsibility Norms as per the Rules and Provisions under the Company Act and revised from time to time.
Labour Laws	Comply with the Indian Labour Law (for establishments in India) or International Labour Organization (for international establishments), as specified within the provisions of the various Rules and Regulations prepared from time to time.
Employee Well-being and Gender Inclusivity	Should have internal policies and guidelines to promote employee and staff wellbeing.

Draft specifications for AC Manufacturers based on parameters requested on the GeM portal

Conformity / Certification

BEE Star Rating	5	
Conformity to Indian Standard	IS 1391 latest	

Capacity / Generic

Type of Air conditioner	As per buyer requirement	
Technology of AC	Inverter (Variable Speed)	
Nominal cooling capacity in Ton / (kcal/hr)	1 Ton / 1.5 Ton	
ISEER	Based on capacity (ISEER specs to be confirmed)	New parameter to be added on GeM portal.
Coil Material	As per buyer requirement	
Eco-friendly refrigerant	Yes	
Type of refrigerant	Selection based on GWP and ODP values of refrigerants.	New parameter to be added on GeM portal.
Minimum length of copper pipe and suitable connecting electrical cable for installation and commissioning	As per buyer requirement	
Packing List	As per buyer requirement	
Any other Features	As per buyer requirement	

Warranty

Warranty on Machine	As per buyer requirement	
Warranty on Compressor	As per buyer requirement	

Installation

Installation and commissioning	As per buyer requirement	
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Total cost of ownership (sample calculations for 1.5 TR)

Room Air conditioner	Initial Cost (INR) approx. (A)	Annual Electricity Consumption (kWh/ year) (B)	Annual Cost of Electricity (INR) (C) = B x Cost of electricity.	Approx. Annual Maintenance Costs (INR) (D)	Total Operational Use Cost (over lifetime) (INR) (E) = (C+D) x No. of years	End of Life (Disposal Costs in INR) (F)	Total Cost of Ownership (INR) (G) =A+E+F
3 Star (3.5 ISEER)	34,600	1,125	11,250	1,500	89,250	2,500	126,350
5 Star (4.5 ISEER)	45,734	880	8,800	1,500	72,100	2,500	120,334
Green RAC (5.33 ISEER)	50,000	766	7,660	1,500	64,120	-2,500 (Buy-back)	111,620

Assumptions:

1. The calculations are done for a room air conditioner (split, variable speed) of 1.5 TR.
2. The initial cost is an approximate average based on study of products available in the market. End of life disposal cost is also included here. In a green RAC, this can be considered as an income with introduction of the buy-back clause.
3. This assessment has been done for variable speed room air-conditioner.
4. The annual electricity consumption is based on a study of the range of the products listed on the BEE website. The tariff is considered as INR 10 /unit cost of electricity. The annual cost will vary based on electricity tariff structure in different cities as well as climatic conditions.
5. The total cost of ownership is based on service life of 7 years.

Environmental Cost Evaluation based on GHG emissions (kg CO₂ emissions)

Room Air conditioner	Annual Electricity Consumption (kWh/ year)	Life-time Electricity Consumption (kWh)	GHG emissions over life-time (kg CO ₂ emissions)	GHG emissions saved over life-time by 1 RAC (kg CO ₂ emissions)
3 Star (3.5 ISEER)	1,125	7,875	6,458	
5 Star (4.5 ISEER)	880	6,160	5,051	1,406
Green RAC (5.33 ISEER)	766	5,362	4,397	2,061

Assumption: 1 kWh = 0.82 kg CO₂ (Energy from Source for Indian Grid, CEA)

Benefits of Green RAC specifications

Proposed Green RAC specifications	Benefits
Type of air conditioner – Fixed speed air conditioners to be discontinued.	Inefficient air conditioning products are not included in the list. The total cost of ownership of energy-efficient products is less when compared with inefficient products.
Energy Efficiency criteria for room air conditioners.	Lower environmental impact, during the use phase. The total cost of ownership of products with higher ISEER is less when compared with products with lower ISEER.
Criteria for refrigerant (GWP) has been specified.	Lower environmental impact throughout the entire life cycle of the product, including end of life disposal.
Recycled plastic in components and “take-back” clause in contracts.	Less burden on the use of virgin material in manufacturing and reduced environmental impact at the end-of-life stage due to safe disposal.
Sustainable packaging criteria	Less burden on the use of virgin material.